MIL-F-16377/23A(SH)
9 May 1983
SUPERSEDING
MIL-F-16377/23(SHIPS)
30 November 1970

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, GENERAL LIGHTING

200 WATTS, 115 VOLTS, WATERTIGHT, SYMBOLS 57, 57.1, 69, 69.1 AND 103

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377.



SYMBOLS 57 AND 69.1



SYMBOLS 57.1 AND 69



SYMBOL 103
(SHOWN WITHOUT SHIELD)

FIGURE 1. Typical view of Symbols 57, 57.1, 69, 69.1 and 103.

All dimensions in inches.
For tolerances other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

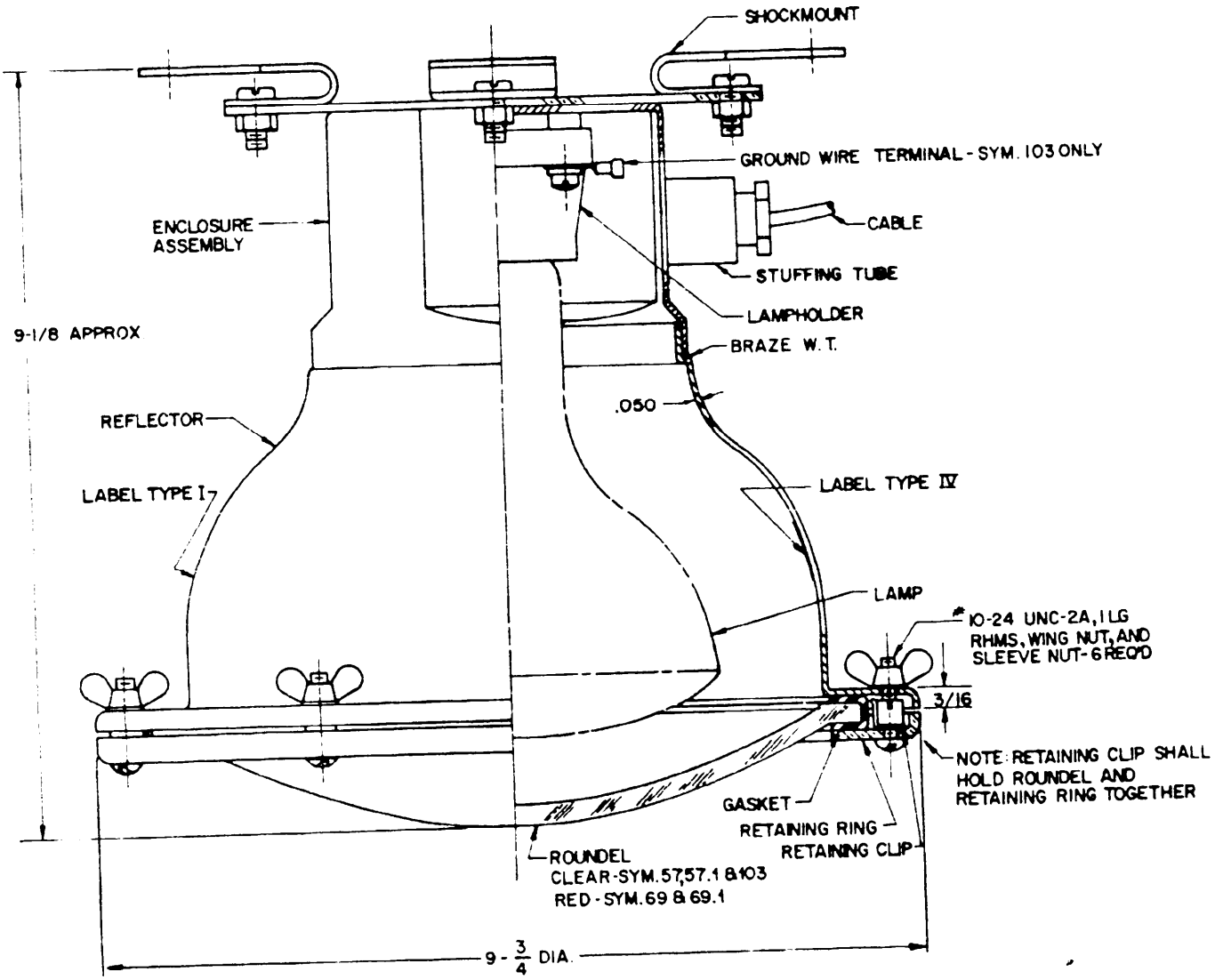


FIGURE 2. Fixture assembly, Symbols 57 and 69.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

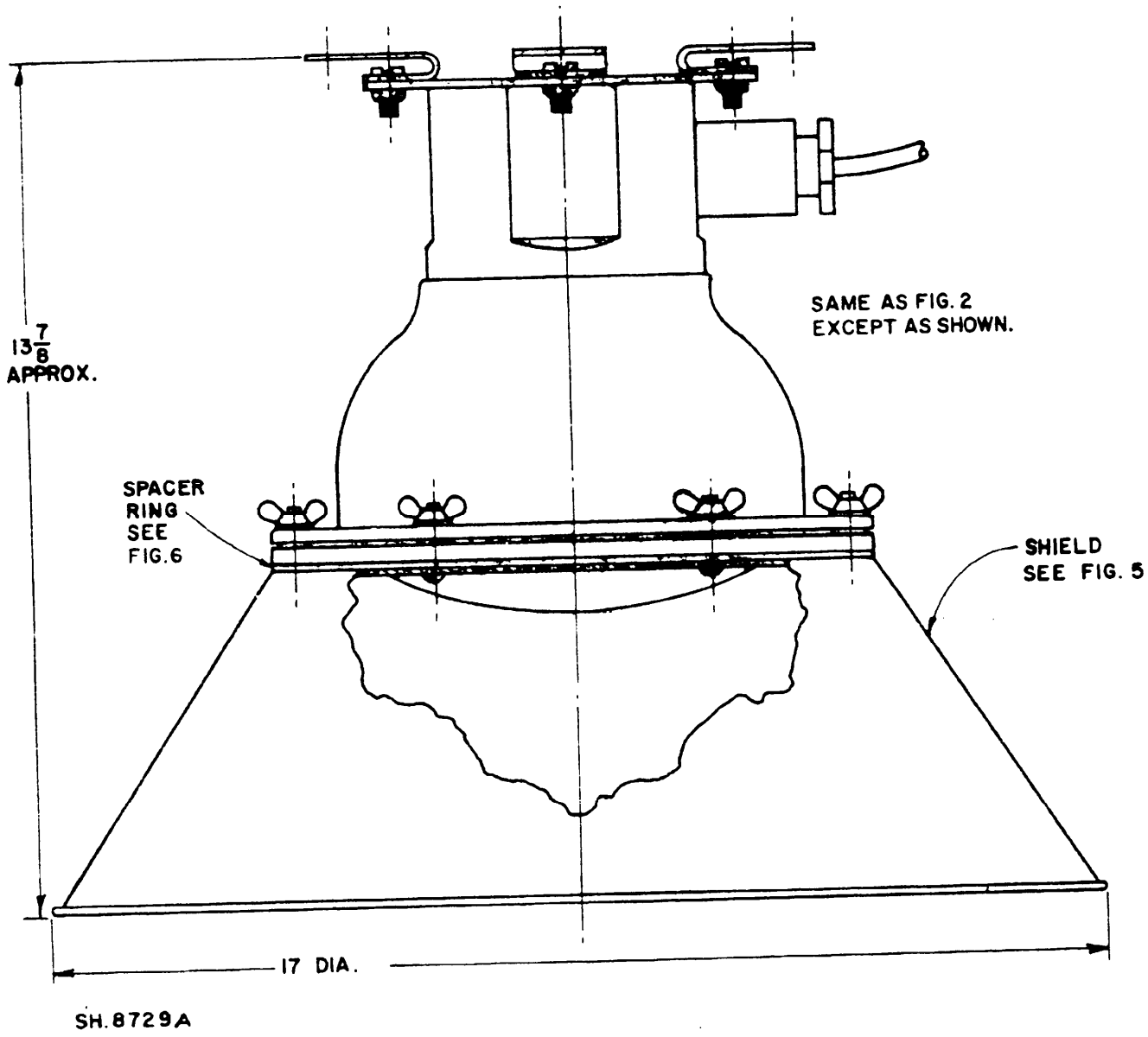


FIGURE 3. Fixture assembly, Symbols 57.1 and 69.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

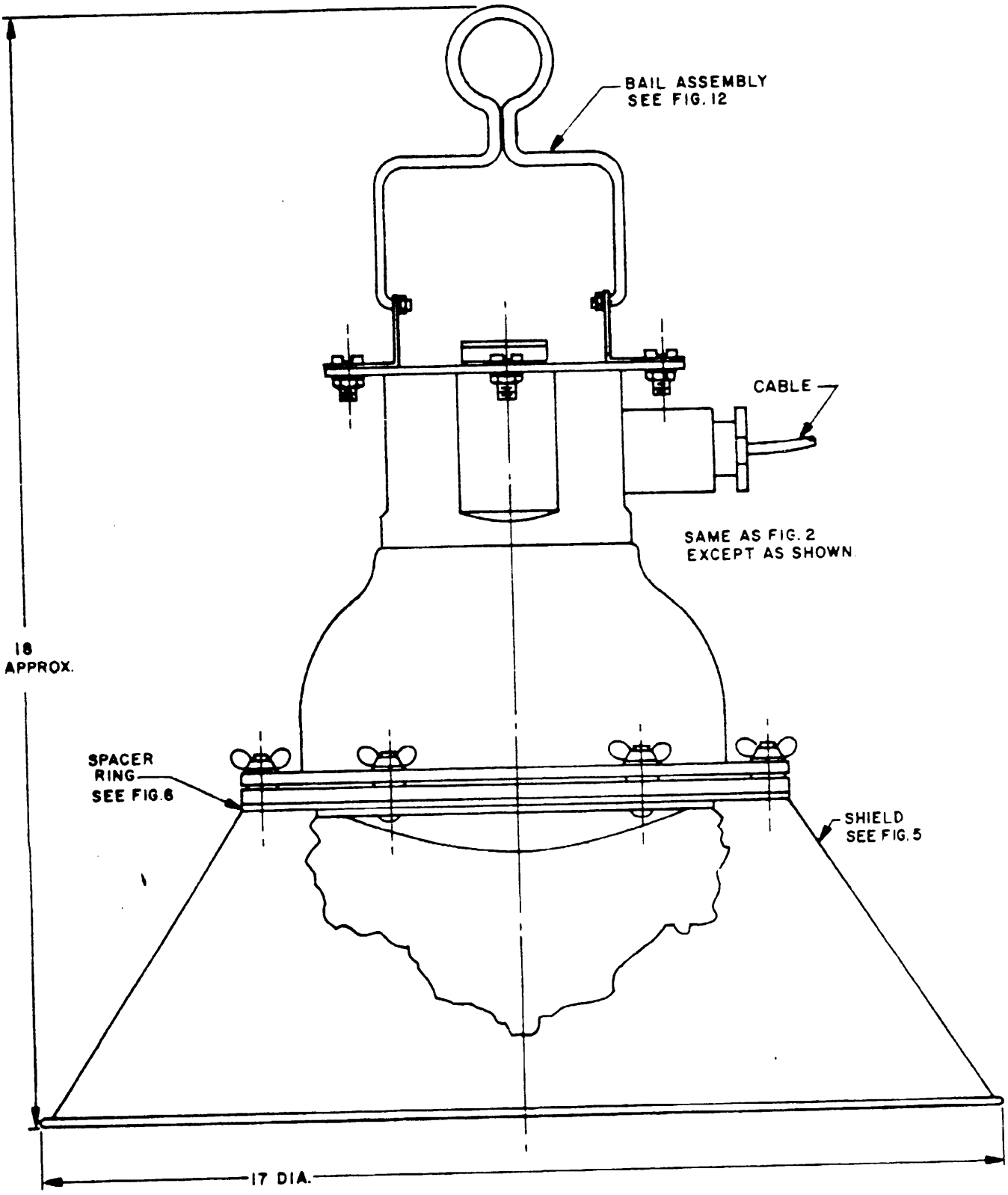


FIGURE 4. Fixture assembly, Symbol 103.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

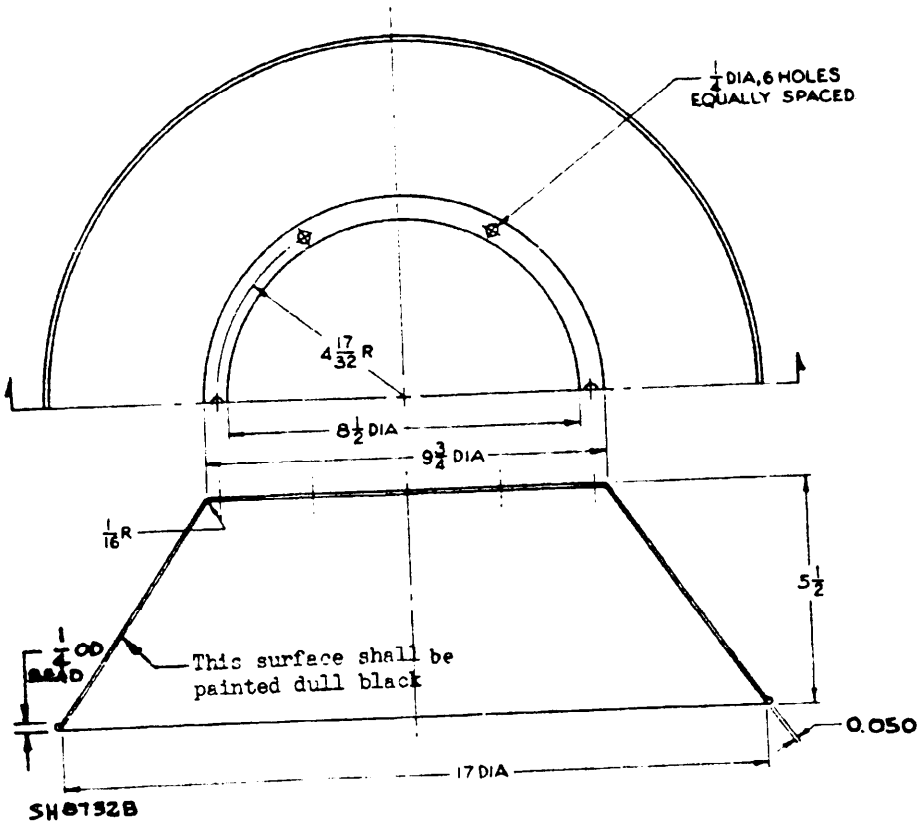
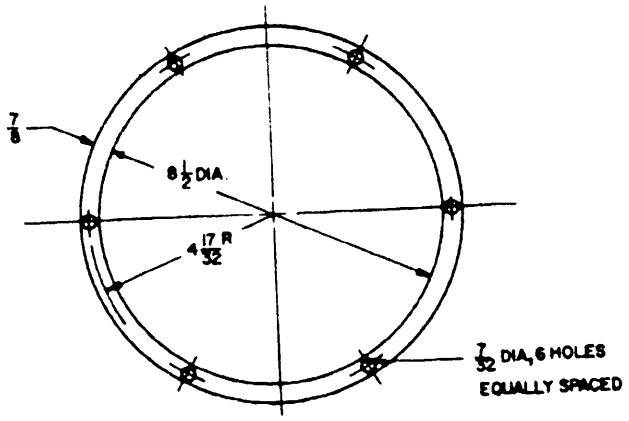


FIGURE 5. Shield.



MATERIAL: FIBER, 1/32 THICK

FIGURE 6. Spacer ring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

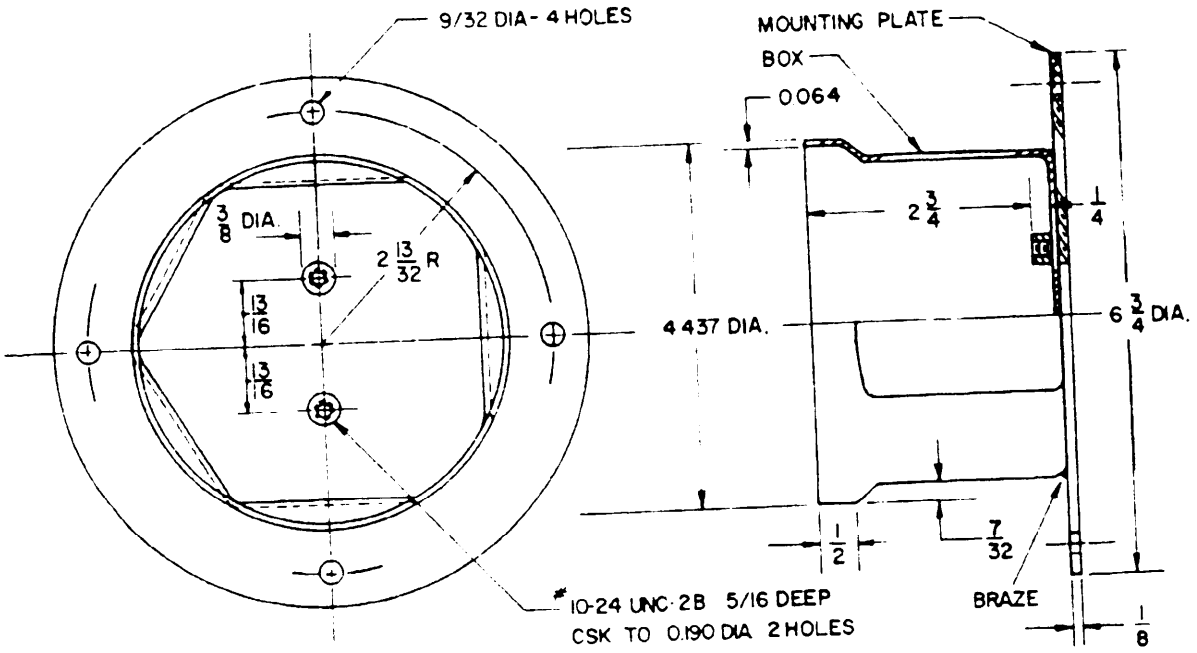


FIGURE 7. Enclosure.

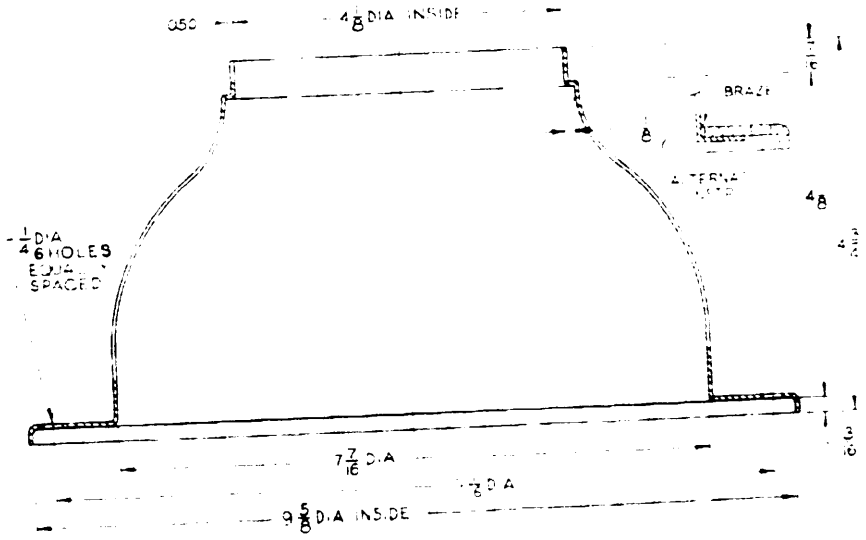


FIGURE 8. Reflector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

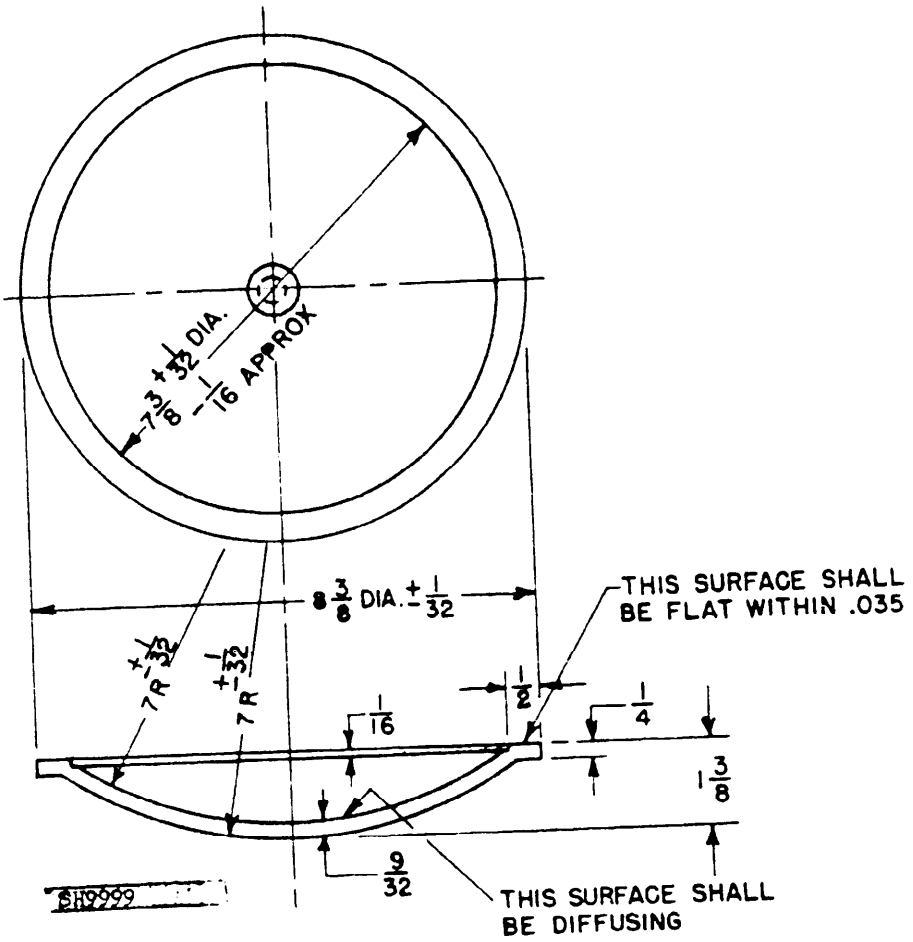


FIGURE 9. Roundel.

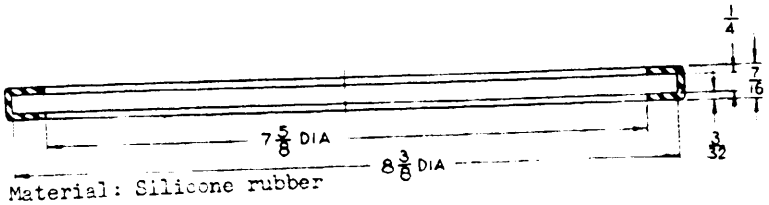


FIGURE 10. Gasket.

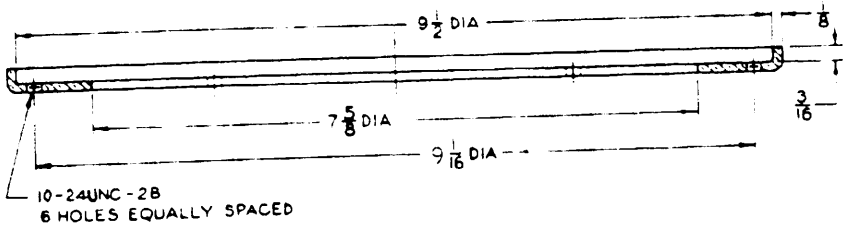


FIGURE 11. Retaining ring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

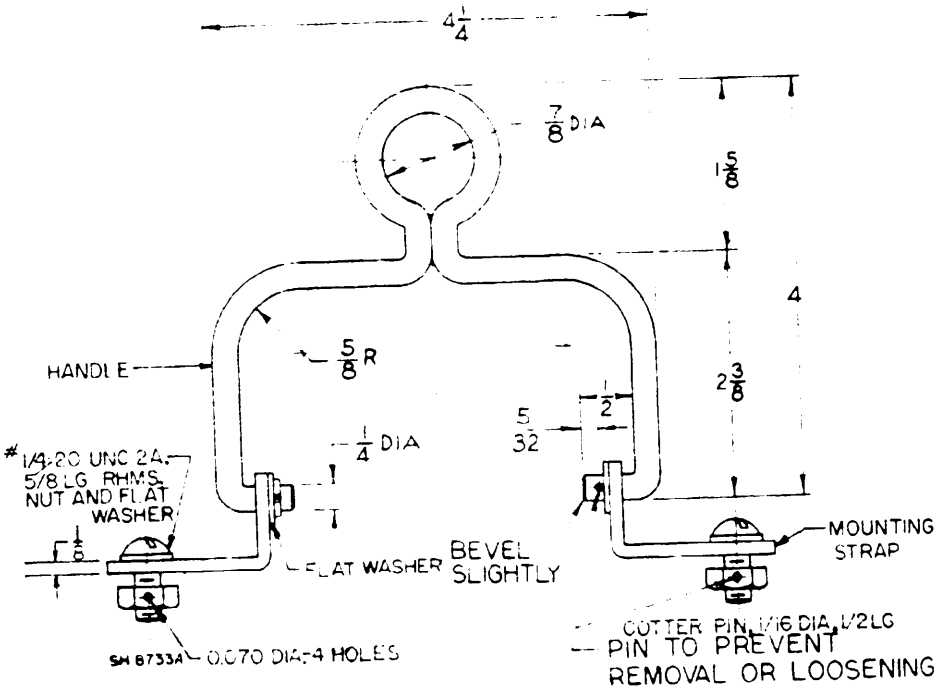


FIGURE 12. Bail assembly.

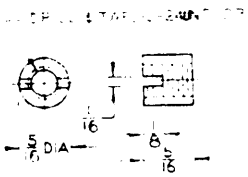


FIGURE 13. Sleeve nut.

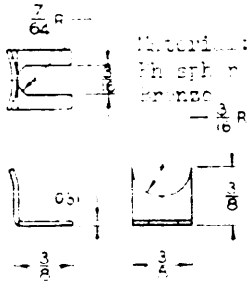
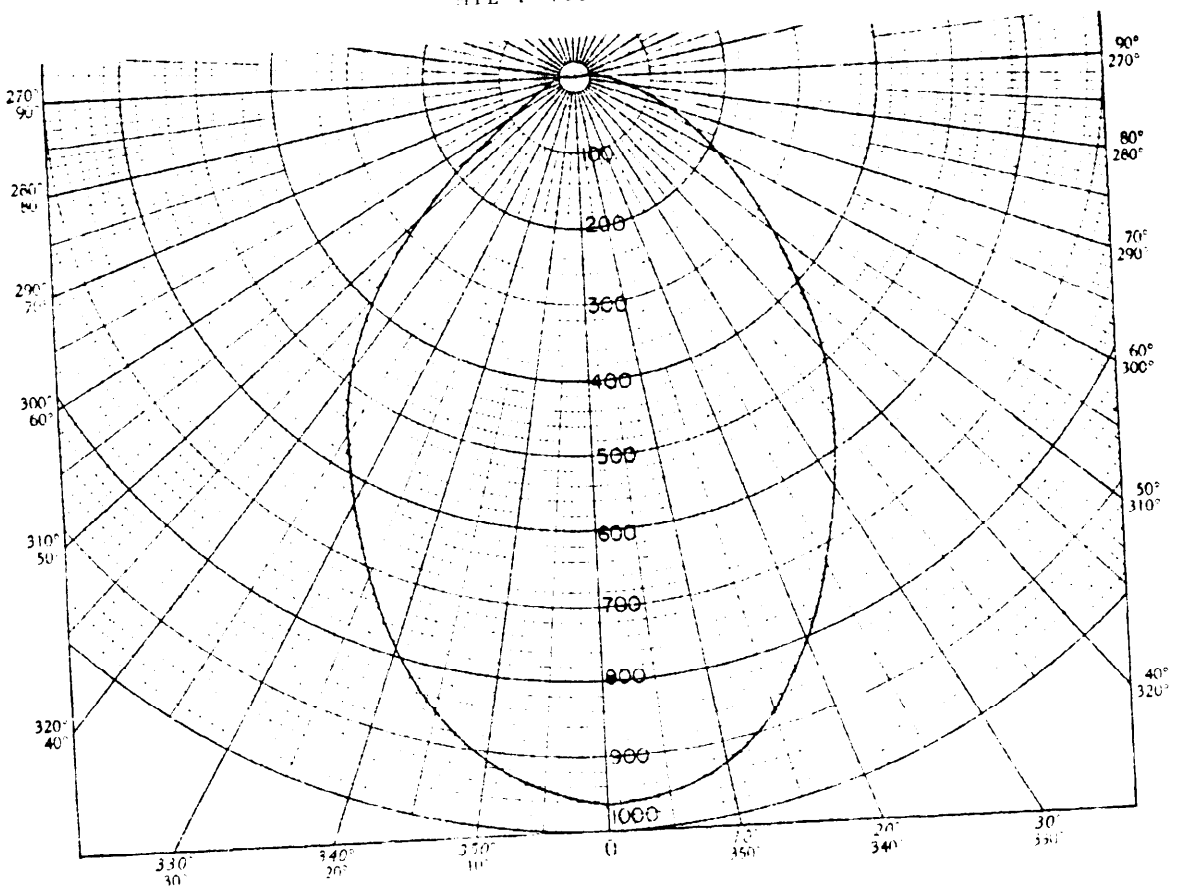


FIGURE 14. Lens retaining ring.



FIGURE 15. Mounting strap.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.



ANGLE	CANDLEPOWER SYMBOL	
	57, 103	69
0	967	
5	951	
10	910	
15	860	
20	787	
25	725	
30	640	
35	562	
40	460	
45	377	
50	300	
55	230	
60	180	

ANGLE	CANDLEPOWER SYMBOL	
	57, 103	69
65	133	
70	115	
75	77	
80	60	
85	12	
90	27	
95	14	
100	10	
105	6	
110	0	
115		
120		
125		

Photometric test data:

Distance: 10 feet.
Voltage: 115 volts.
Ambient temperature: 76°F.
Lamp: 200 watts, rough service conforming to MS15543-2, furnished by laboratory for test, shall be used.

FIGURE 1b. Candlepower distribution curves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

CLASSIFICATION:
Type: II
Class: 2

SYMBOL NUMBERS
57 and 57.1: Fixture, clear transparent diffusing roundel, permanently mounted, for use with 200 watt incandescent lamp.
69 and 69.1: Fixture, red transparent roundel, permanently mounted, for use with 200 watt incandescent lamp.
103: Fixture, clear transparent diffusing roundel, portable, with bracket, for use with 200 watt incandescent lamp.

DESIGN:
Electrical rating: 200 watts, 115 volts.
Dimension and configurations: See figure 1 through 15.
Mounting dimensions: See figure 28 of MIL-F-16377.
Photometric data: See figure 16 (Symbol 57, 57.1 and 103 only).
Material: Unless otherwise specified, material shall be brass.
Roundels:
Symbols 57, 57.1 and 103: Clear transparent diffusing, see figure 9.
Symbol 69 and 69.1: Red transparent, Part number MS24489-1 of MS24489.
Cable:
Symbol 57, 57.1, 69, and 69.1: 10 feet of type DSGU-3 of MIL-C-915/29.
Symbol 103: 15 feet of type TSGU-3 of MIL-C-915/30. Ground connection shall be the red lead.
Lampholder: MIL-L-9470/6, two contact.
Shockmounts: MIL-F-16377/41, Part Number M16377/41-029.
Enclosure: Watertight
Stuffing tube: Shall be symbol 2446.1 of NAVSHIPS drawing 803-1197161.
Stuffing tube packing: Shall be Symbol 2447.2 of NAVSHIPS drawing 803-1197101 except that material shall be silicone rubber, capable of withstanding constant temperature inside the fixture of 170° C.
Weight:
Symbol 57 & 69.1: 9 pounds, 0 ounces max.
Symbol 57.1 & 69: 11 pounds, 8 ounces max.
Symbol 103: 12 pounds, 8 ounces max.
Lamp: These fixtures shall be designed to accommodate a 200-watt, 115 volt, R-40, reflector flood, medium screw base, rough service lamp conforming to MS15543-2, Industry number 200RL/RS/115V. (Lamp not furnished).
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figure 2.

(a) One type I.
(b) One type IV. Apply one type IV label with the following information in 3/16 inch letters: "USE THE FOLLOWING REPAIR PARTS ONLY: LAMP NSN6240-00-248-4280, WHITE LENS NSN6210-00-299-2893, RED LENS NSN6210-00-633-6886."

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixture	See table I
Repair parts:	See table II
Quality conformance:	
Comparison inspection:	Same as the first article inspection
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.

TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	Symbol 57, 57.1, 69 and 69.1
Vibration	3.5.13	4.8.8	Symbol 57, 57.1, 69 and 69.1
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Grounding circuit	3.6.13.3	4.8.11	Symbol 103 only
Continuity of grounding	3.6.13.4	4.8.12	Symbols 57, 57.1, 69 and 69.1
Leakage current	3.6.13.5	4.8.13	Symbol 103 only
Optical uniformity	3.7.4.1.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	6 ft. pounds
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.7	4.8.6.3	Symbols 57, 57.1 and 103
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbols 69 and 69.1
Light output	3.7.1	4.8.6.1	Symbols 57, 57.1 and 103
Magnetic permeability	3.4.8	4.8.18	
Salt spray	3.3.15	4.8.10	

TABLE II. First article and comparison inspection for repair parts.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	1/
Mechanical shock	3.7.5.6	4.8.17.3	1/ 6 ft. pounds
Thermal shock	3.7.5.7	4.8.17.4	1/
Breakage pattern	3.7.5.9	4.8.17.6	1/
Color and luminous transmittance	3.7.3.1	4.8.6.3	1/ Red
Color and luminous transmittance	3.7.3.7	4.8.6.3	1/ Clear

1/ Roundels only.

TABLE III. Part number and general information.

Military 1/ part number M16377/23-	Item Name	National Stock Number (NSN)	Remarks
57	Complete fixture, Symbol 57	6210-00-283-9691	
57.1	Complete fixture, Symbol 57.1		2/
69	Complete fixture, Symbol 69		3/
69.1	Complete fixture, Symbol 69.1		4/
103	Complete fixture, Symbol 103		
001	Roundel, clear transparent	6210-00-299-2893	5/
002	Roundel, red transparent		
003	Bail assembly	6210-00-399-7120	
004	Shield and spacer assembly	6210-00-548-0107	Figures 5 and 6
005	Casket	5930-00-039-4228	Figure 10

1/ Formerly covered on NAVSHIPS Drawing 9000-6401-74220 and Military Standard MS17276.
2/ Symbol 57.1 is made up of Parts M16377/23-57 and M16377/23-004.
3/ Symbol 69 is made up of Parts M16377/23-57 (less M16377/23-001) and M16377/23-002.
4/ Symbol 69 is made up of Parts M16377/23-69 and M16377/23-004.
5/ NSN 6210-00-548-0228 for M16377/23-002 has been superseded by NSN 6210-00-633-6886 of MS24489-1.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing Activity:
NAVY-SH
(Project 6210-N564)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:44Z
Check the source to verify that this is the current version before use.